

Design Microsoft Power Platform solutions by using AI-enabled tools

1. Analyze requirements to identify components and options for implementation

- Dataverse is the data store for the Power Platform.
 - You can create tables which includes columns and rows, and business rules and more.
- Canvas Power Apps allow you to create pixel-perfect apps.
 - It can use data from the Dataverse and elsewhere from over 1,000 connectors.
- Model-driven Power Apps concentrate on the data using forms (information about a single row) and views (multiple rows).
- The AI hub analyses data, extracting or analysing information from documents, text, structured data and images, and create prompts.
- Copilot Studio allows for the creation of chatbots and agents, which can run other agents.
- Power Automate allows the running of flows, which are run based on triggers and can perform actions.

2. Evaluate built-in agents to include in the business solution

- You can use plans to create a business solution.
- To do this, go to Plans (on the left-hand side) and go to "Create a plan".
 - Then describe your business problem - for example "Equipment request and manager approval tracker. I also need an analysis per month. I also want to be able to discuss the equipment ordered".
 - You can also attach a file to it.
- Once a plan is created, at least in part, you can save the plan and click on the Share plan icon. You can then add users, together with their permission:
 - Viewer - to view the plan only, or
 - Co-owner, to view, edit and share the plan, but not delete the plan or change owners.

2. Evaluate built-in agents to include in the business solution

- You can also click on "Manage access" to manage existing access, or "Copy link" to copy to the clipboard a link to the plan.
- Once you have added users, you can click Share.
- You can also click on Share - Export as a PDF, which can be used to share with others who do not have access to Power Apps.
- Once a plan is created, it can be accessed using Plans (on the left-hand side).
 - If you access it again, it will open in Read-only mode. In this mode, you can view the plan, but not make any changes. You are also not able to Share it with other users (but you can Export it as a PDF).
- It will then hand it over to the Requirement Agent.
 - It breaks down the business problem into individual needs.
 - You can then click on Edit, which allows you to:
 - Edit the user roles,
 - Clicking the ... next to the role allows you to add or delete roles.
 - Edit the text, including adding or deleting user requirements.
 - Use Copilot to describe changes.
 - Once you have completed any changes, click on "Looks good".
- It then hands it onto the Process Agent.
 - This suggests processes to solve the needs.
 - You can click on Edit, then "View process" to:
 - see steps (or activities or tasks - rounded rectangles), events (circles) and decision points (or gateways - diamonds - these can be Yes/No decision, concurrent paths, and conditional paths), together with the user role.
 - You can alter the text or delete the step, event or decision point.
 - You can also connect two objects together by dragging from one node to another.
 - You can add activities, events or gateways.
 - You cannot change a node type - you would need to delete and recreate it.

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2. Evaluate built-in agents to include in the business solution

- Once you click on "Looks good", it is then the turn of the Data Agent.
 - It creates the data model with tables and suggested columns.
 - You can click on the "Show details" icon to open the tables in the Data Workspace.
 - You can click on the ... next to a table to:
 - Edit table in the Data Workspace,
 - You can also click on Tables on the left-hand side of the screen.
 - Replace with existing table from the Dataverse,
 - Remove from plan.
 - It also says at the bottom the name of the solution that the tables will be saved in.
 - You can click on "Change solution" to change the solution to:
 - A new solution, and optionally set it as the preferred solution, or
 - Click Advanced to set an existing solution,
 - You can also click Advanced to select an existing publisher.
- You can also see a list of the objects in the plan by clicking on "Objects" on the left-hand side.
- After clicking "Looks good", the Solution agent then suggests the applications, flows and other objects to be used, including:
 - Canvas or model-driven apps,
 - Power Pages websites,
 - Power Automate flows,
 - Power BI Report, and
 - Copilot Studio Agents.
- You can then:
 - Click on Edit,
 - Click Replace to replace the proposed item with an existing app. Note: it will not be included in the new solution.

3. Recommend extensibility options

- Click on "Add technology" to add additional items,
 - Click "Save tables",
 - Give the solution a name, or select an existing solution,
 - Select a publisher, and
 - Click Save.
- It is possible to create a plan from an existing solution.
 - Go to Solutions (on the left-hand side), check the solution and click "Create a plan".
 - Your solution must have at least one app and one associated table.
 - This can be used for documentation purposes, and to help improve an existing solution.

3. Recommend extensibility options

- Extensibility options in Power Platform include:
 - Power Automate (see topics 31-36) – you can create workflows that can automate processes:
 - Use triggers which run on changes in data or specific events, and actions like sending emails, creating rows or calling external services to automate processes.
 - You can integrate services such as Azure Functions, which allows you to write custom code in C#, JavaScript and Python.
 - You can write business rules, to implement and enforce business logic at the data layer (see topic 45).
 - You can also create business process flows, to guide users through a series of steps (see topic 46).
 - You can also use Modern commanding to extend your model-driven apps
 - Click on the ... next to a view in the model-driven app designer, and go to "Edit command bar".
 - Power Fx allows you to use Excel-like formulas, allowing for a low-code environment.
 - You can use this in Modern commanding, and in canvas apps, among other places.

4. Recommend environment types

- Create custom connectors.
 - This allows you to talk to external data sources and services - for example, Azure Functions. It uses REST API.
- Build custom User Interface components using the Power Apps Component Framework
 - This uses languages like C# to create your custom code, and then package the component into a solution file.
- Build plug-ins which can customize Dataverse data processing
 - You would write custom business logic, which can run based on Dataverse events ("messages") happening, either:
 - Before the operation is validated, so it can be cancelled at this stage,
 - Before the operation occurs, or
 - After the operation occurs.
- If you click on the form (instead of a column), you can also go to the Properties pane - Events tab. Events can also be created:
 - The events are
 - Form OnLoad and OnSave,
 - Tab TabStateChange (when a tab is expanded/collapsed).
 - Field OnChange (data in a field changes and the control loses focus),
 - iFrame OnReadyStateComplete (iFrame content loads).
 - The event is a function which executes code in a JavaScript web resource

4. Recommend environment types

- Environments allow you to store, manage, and share within the Power Platform:
 - Data, Apps, and Flows.
- Each environment allows one Dataverse database.
- An environment is created:
 - By going to Power Apps – Gear – Power Platform Admin Center – Environments – New.

4. Recommend environment types

- Enter a new and select Type (see next page).
- Select the Region
- Create a Dataverse database, if wanted.
- In an Azure Active Directory (Azure AD) tenant.
- Two built in roles are:
 - System administrator role or Environment Admin role, which allows for:
 - Add/remove users/groups from Environment Admin or Maker roles.
 - Provision a Dataverse database.
 - View/manage all resources in the environment.
 - Set data loss prevention policies.
 - Environment Maker role, which can be used for:
 - Creating resources within an environment (such as apps, connections, gateways, and Power Automate Flows).
 - Distribute their apps to other users in an organization.
 - This role does not have automatic access to a Dataverse database.
- Multiple environments can be used for:
 - Different geographic locations.
 - Different geographic areas have different laws for data residency. For example, the European Union has the General Data Protection Regulation.
 - Development/testing/production.
 - Departments.
- Data access (to the Dataverse, for example).
- There are six types of environments:
 - Default
 - Environment created automatically to the tenant.
 - Meant for experiments, exploring, and lightweight app trials.
 - Cannot be deleted, backed-up, or restored.
 - No users will be automatically added to Environment Admin role.

5. Apply a Microsoft Power Platform solution and ALM strategy

- All licensed users will be automatically added to the Environment Maker role to this environment.
 - Admins cannot restrict access to the default environment.
 - Limited to 32 Gb of storage capacity.
 - You may wish to rename it “General”.
- Production
 - For permanent work. Need 1GB available database capacity.
- Sandbox.
 - Development and testing. Useful for testing updates to applications.
- Trial (standard)
 - Short-term (expire after 30 days).
- Trial (subscription-based)
 - This is for multiuser, multi-department trials, aimed at multiple users. It has an end date which can be extended.
- Developer
 - Developer Plan license – only intended for use by the owner. You cannot use security groups.
- Microsoft Dataverse for Teams.
 - Automatically created in Teams.
- You can create a Managed Environment.
 - This allows admins to ability to manage this, and other, environments collectively.

5. Apply a Microsoft Power Platform solution and ALM strategy

- Used to transport apps and components from one environment to another.
- Contains site maps, tables, processes, web resources, flows and more.
- By default, all objects will be added into the "Common Data Services Default Solution".
- This is not the best, because all objects by all makers will be stored there, not just yours.

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5. Apply a Microsoft Power Platform solution and ALM strategy

- To create a solution:
 - Go to Solutions on the left-hand side.
 - Go to “New solution” at the top.
 - Enter Display Name, Name, Publisher, and version.
 - It will be an unmanaged package.
- To change the solution that your new objects will be stored in:
 - Go to the solution,
 - Click on "Set preferred solution".
- There are two different types of solutions:
- Unmanaged solutions
 - Used in development environments.
 - There are two default unmanaged solutions:
 - Common Data Service Default Solution. It has a random publisher prefix.
 - Default Solution. All components in the system. It cannot be Exported.
 - These two unmanaged solutions use the default publisher and publisher prefix assigned to the solution.
 - If an unmanaged solution is deleted, it is only the container which is deleted. All unmanaged customizations then belong to the default solution.
- Managed solutions
 - Used to deploy to other (non-development) environment – test, User Acceptance Testing (UAT), system integration testing (SIT) and production.
 - Managed (“controlled” in other programs) solutions should be generated by exporting an unmanaged solution as managed.
 - Components cannot be directly edited within a managed solution.
 - You could add them to an unmanaged solution, which creates a dependency. Managed solutions with a dependency cannot be uninstalled.

5. Apply a Microsoft Power Platform solution and ALM strategy

- You can also lock down components using going to Power Apps/Automate – Solutions – NameOfSolution – NameOfComponent – Managed properties.
- Managed solutions cannot be exported – only deleted. When deleted, all components as well as the solution will also be deleted.
- The Default solution:
 - Has a random customization prefix.
 - It is added to all items, columns, relationships or choices in the default solution.
 - You cannot change the publisher or prefix of a default solution.
 - Cannot be exported.
- To create a solution, go to Power Apps – Solutions – New solution.
 - Add the display name, name, publisher, version, and optional description.
 - The Name is a unique solution name, and cannot be edited after changing the solution.
 - New publishers will include Display Name, Unique Name, Publisher Prefix, Option Value Prefix and Contact Details.
 - Option Value prefix is for naming “Choices”, and is a number between 10,000 and 99,999.
 - To add new apps, flows and tables, go to Power Apps – Solutions – NameOfSolution – and click on “New”.
 - To add existing apps, flows and tables to an existing solution, go to Power Apps – Solutions – NameOfSolution – and click on “Add existing”.
 - You may be able to select from the “From solutions” and “Outside solutions” tabs.
 - You cannot add instant flows created outside of a solution where the flow trigger is set to manual.
 - To edit them, select the apps, flows and tables in Power Apps – Solutions – NameOfSolution – and click on the “Edit” (or go to the ... and click on Edit).
 - The details button for flows only will take you into the Flows – Details section.

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5. Apply a Microsoft Power Platform solution and ALM strategy

- This will not import/export data. To do this, go to the Table, and then:
 - Import – Import data with Dataflows,
 - Import – Import data from Excel,
 - Export – Export data (so you can edit it in Excel)
- The default solution cannot be exported.
- To export a solution, go to Power Apps/Automate – Solutions, select the solution and go to Export.
 - You should click “Check for issues” before you export.
 - If you have made changes that you would like to export, click “Publish” before you export.
 - You can then give the export a version number, and whether it is unmanaged (development or source control) or managed (test or production).
 - The version number is four numbers separated by dots. Those numbers are major number, minor number, build number and revision number.
- It will take a little while, but a zipped file will be downloaded to your computer.
 - The file will have the solution name, managed/unmanaged, and version number.
- You can then Import it into your other environment by going to Power Apps – Solutions.
 - You cannot add instant flows into a solution when:
 - it was created outside of a solution and
 - the flow trigger is set to manual.
 - You also cannot add custom connectors created outside solutions.
 - You can “View solution package details” if you would like.
 - If the “solution package contains an update for a solution that is already installed” (which you will see in a yellow bar), you can upgrade (default) or click on Advanced Settings (more in “Support deployment”).
 - You can have “Enable SDK messages and flows included in the solution” – these are external packages.

6. Create and edit tables in the data workspace

- If your solution contains connection references, you will be prompted to select or create the connections.
- If your solution contains environment variables which are not already present, you will be prompted to enter values.
- If your import was not successful, you can click “Download Log File” – it will probably show that some required components were not included in the solution.

Build data models

6. Create and edit tables in the data workspace

- You can create tables in the data workspace by going to:
 - Create (on the left-hand side) - Start from data - Create new data,
 - Tables (on the left-hand side) and then
 - "New tables" - "Create new tables", and either "Start with Copilot" or "Start from blank",
 - "Get started with Copilot" or "Start with a blank table".
- If you use Copilot, then:
 - You can describe what you want to create,
 - You can optionally click on the Settings button and select:
 - Multiple/One table,
 - Table size: Small, Medium or Large, and
 - Include relationships.
 - Click Generate, and it will create a draft structure which can then be edited.
- In the Command bar, you can click:
 - Back - to leave the Data Workspace,
 - New table - you can:
 - Describe new tables (using Copilot),
 - Add columns and data - this creates a blank table,

6. Create and edit tables in the data workspace

- Start with Excel or a .CSV (comma separated values) file, or a SharePoint list
- Add existing table from the Dataverse.
 - You can narrow it down by turning on "Only in current solution".
- View/hide data - this opens a pane at the bottom which shows sample data,
- Create relationships - this allows you to create a relationship between two tables. You select:
 - Relationship table - either One-to-many or Many-to-one (not Many-to-Many or One-to-one),
 - The key columns between the tables, and
 - The Lookup column in the "One" table,
 - Whether data is required for the row to be saved,
 - Remove - deletion of the table and data.
 - Alternatively, you can drag between two +s at the bottom of tables to create a relationship.
- At the bottom-left hand corner, you can:
 - Zoom in/out,
 - Click "Fit view" to view everything, and
 - Click Show/hide map to show an overview of the entire canvas.
- You can click on the ... next to a table and click:
 - View/hide data,
 - Row ownership - either:
 - Rows owned by a user or team, or
 - Rows owned by an organization
 - Properties - you can change:
 - The Display and Plural name (required),
 - Description,
 - Primary column - a text column less than 850 characters,

6. Create and edit tables in the data workspace

- If you click on "Advanced options", you can also edit the (internal) schema name.
 - You can then click Save to save these changes.
- Remove - to delete the table and data.
- In the Table row editor, you can:
 - Add new rows or new columns,
 - Change the "Row ownership",
 - Click on Properties,
 - Bring the Table row editor to occupy the entire window.
 - Click on a column to:
 - Edit column to edit:
 - Display name,
 - Data type,
 - At the time of recording, you cannot use Rich text, Customer, Autonumber or Formula data types.
 - You would have to create columns with those data types outside of the Data Workspace.
 - Format,
 - Whether it is required (a * will be next to the column if it is required),
 - If you click on "Advanced options", you can also edit the (internal) schema name.
 - Insert new column,
 - Delete existing column (apart from the primary key).
- You can also use the Copilot pane to alter the tables. You can either enter the prompt and click the arrow, or go to "View prompts", and go to:
 - Create a table(s)/rows/columns/relationship,
 - Import Excel/.CSV file or SharePoint list,
 - Rename selected table/column,
 - Change column/relationship type,

7. Create and modify standard tables

- Delete table/column/relationship, or
- Suggest table/column/relationship.
- When you have finished, click "Save and exit".

7. Create and modify standard tables

- To create a table in the Power Apps portal (make.powerapps.com), go to Tables, and then click on "New table – New table".
- There are five types of tables:
 - Standard – base set of entities created in the Dataverse.
 - Fields can be added, but not deleted.
 - Display names can be renamed.
 - Tables can be hidden (using security).
 - Custom – a "blank" table.
 - Complex – contains server-side business logic. May contain workflow and plug-ins.
 - Requires Dynamics 365 license or "per user" plan.
 - Restricted – stores data not normally used outside of Dynamics 365, or maintains data in a specific way.
 - Requires Dynamics 365 license.
 - Virtual tables – a table from an external source, such as Azure SQL Database.
- Activity tables are for tasks. They have columns such as:
 - time dimensions (start, stop, due data, duration),
 - Subject, description,
 - Status: opened, canceled, completed.
- Standard activity tables include:
 - Appointment – Commitment representing a time interval with start/end times and duration.
 - Email – Activity that is delivered using email protocols.
 - Fax – Activity that tracks call outcome and number of pages for a fax and optionally stores an electronic copy of the document.

8. Configure table properties

- Letter – Activity that tracks the delivery of a letter. The activity can contain the electronic copy of the letter.
- Phone Call – Activity to track a telephone call.
- Recurring Appointment – The master appointment of a recurring appointment series.
- Task – Generic activity representing work needed to be done.
- When creating a table, enter:
 - Display name (and plural), and
 - Primary Name column (only one per table – not compound).
 - This is then automatically created.
 - There is also a hidden Primary Key, a GUID.
 - You can have alternative keys (index) – this can be single or compound, and will be Unique and Required.
 - To do this, go to Power Apps – Data – NameOfTable – Keys – Add Key.
 - Already inserted data needs be unique, or the Key will not be inserted.
 - You can have up to 5 Keys per table.
 - Keys, as an index, can improve database performance. It is also useful for linking to external data sources.

8. Configure table properties

- Dataverse allows for two types of ownership of entities:
 - Organization owned. Access level is at the user level.
 - Activity tables cannot be owned by an organization.
 - User or Team owned (Teams are owned by a Business Unit). Access level is at the:
 - Organization level,
 - Business Unit level,
 - Business Unit and Child Business Unit level, or
 - User's own records only.

8. Configure table properties

- This happens when the Table is created. It cannot be later changed.
- For this table
 - Apply duplicate detection rules
 - Creates duplicate detection rules.
 - Track changes
 - Find out what has changed since first extracted or last synchronized.
 - Provide custom help
 - Which webpage users will see when they click "Help" in your app.
 - Audit changes to its data
 - Log data creation, changes or deletion, if auditing is enabled for your environment
 - Leverage quick-create forms if available
 - Users can create new rows in a model-driven app by clicking the Create button or "+New".
 - Enable long term retention
 - This would move some data to long term storage
- Make this table an option when
 - Creating a new activity (cannot be turned off when on)
 - Associate activities to rows in the table
 - Doing a mail merge
 - Enable SharePoint document management
 - Allows for the table to integrate with SharePoint
- Rows in this table
 - Can have connections (cannot be turned off when on)
 - Shows how table rows have connections to rows in other tables (which also have connections enabled).
 - Can have a contact email (cannot be turned off when on)
 - An email column will be used as a contact for that row.

9. Create and modify columns

- Have an access teams
 - Creates team templates.
- Can be linked to feedback (cannot be turned off when on)
 - Allow users to write feedback for a row, or rate rows within a defined rating range.
- Appear in search results
 - For example, in Dynamics 365. You may want this table to be hidden from search results.
- Can be taken offline
 - For phone and tablet apps.
- Can be added to a queue (cannot be turned off)
 - May assist with routing and sharing work.

9. Create and modify columns

- To create a column, go to Either New – Column, or click on Columns, and then New column. When creating a column, enter:
 - Display Name,
 - Description (displayed as tooltips in model-driven apps)
 - Data Type
 - Text – Single line of text
 - Plain text. This is a single line of text. Can use Max Length.
 - Text area is for a multi-line text box, up to 4,000 characters. For more, use Multiline Text, up to 1,048,576 characters (but adjustable downwards).
 - Rich text (formatted text).
 - Email – a validated email address.
 - Phone number – a validated phone number.
 - Ticker symbol
 - URL – a validated website.
 - Text – Multiple lines of text

9. Create and modify columns

- Plain text.
- Rich text.
- Number (you can use Minimum and Maximum Values)
 - Whole Numbers
 - Decimal – up to 10 decimal places, exact.
 - Float (floating point) – up to 15 decimal places, approximate, but better performance.
 - Language code
 - Duration – to the nearest minute. Can select from the list or type the number of minutes, or entered as “x minutes/hours/days”, or “x.x hours/days”.
 - Time zone
- Date and time – the Advanced options allow for different Behaviors: User’s local time zone, Date Only and Time zone independent.
 - Date and time, or
 - Date only.
- Lookup
 - Lookup
 - Customer
- Choice
 - Choice [previously called “Option Set”] – list of choices, select only one. Choices [previously called “Multi Select Choice”] – list of choices, select one or more.
 - To create a new Global choice options, click “New choice”. This is reusable and is best when using in multiple columns.
 - ▶ You can edit Global choices by going to Power Apps – Data – Choices (not Tables!).
 - ▶ If you do this, then all columns using this Global choice will be changed.

9. Create and modify columns

- To create a new local choice, click “View more” and choose “Local choice”. Best when using in one column only.
 - When setting up choices, it’s quicker to add “New Items” and then double-Tab through the “New Items”.
 - You can also set up a Default Value.
 - Yes/No [previously called “Two Options”] – list of two choices (not necessarily “Yes/No”), select only one.
 - Currency – choose precision level, or set based on a currency or organization standard.
 - Autonumber – generated by the server.
 - These are really text columns.
 - Have a string or date prefixed number:
 - My-1000, My-1001, My-1002
 - 2025-05-16-1000, 2025-05-16-1001 etc.
 - Or a Custom format
 - CAR-{SEQNUM:3}-{RANDSTRING:6}.
 - SEQNUM gives the Minimum Length (can be exceeded).
 - RANDSTRING Length is between 1 and 6.
 - Needs a starting “seed” value.
 - File
 - File
 - Image – maximum of 1 image column per table, column is always named “tableImage”.
 - Formula (preview)
- Once saved, the data type cannot be changed – you need to delete and add a new column.
 - However, you can rename it by clicking on it.
 - Required (optional, recommended, required),

10. Configure table relationships

- Default value (if required), and
- Searchable (using Advanced Find in model-driven apps using the Web Client – not available in Unified Interface clients).

10. Configure table relationships

- To create relationships:
 - In the Power Apps portal – go to Power Apps – Tables – [Name of Table] – Relationships, and click on “Add relationship”.
- Relationships are links between tables:
 - Typically one-to-many and many-to-many.
 - One-to-many relationships are also called parent-child relationships, with zero, one or many on the “many” side.
 - Usually using a Primary or unique key on the one side.
- In the Advanced options, you can select how changes to one table can affect another one. You can change the “Type of behavior”:
 - Referential. This changes how deletions can be cascaded:
 - Remove Link – Remove the lookup value for all related rows.
 - Restrict – Prevent the primary table row from being deleted when related table rows exist.
 - Parental. Any action on the parent table is also taken on the related table.
 - Custom. You can fine-tune Delete, Assign, Share, Unshare and Reparent:
 - Cascade All – Perform the action on all related table rows.
 - Cascade Active – Perform the action on all active related table rows.
 - Cascade User Owned – Perform the action on all related table rows owned by the same user as the primary table row.
 - Cascade None – Do nothing.
- Once created, a new column of Lookup data type will be added.

11. Configure prompt columns

- Prompt columns allow to generate results based on input columns. These results are stored in your prompt column, and can be used like other data.

11. Configure prompt columns

- The results will be generated asynchronously - they will not be available immediately after data has been added/updated.
- Use cases include:
 - Answering customers questions - when customer questions are stored in Dataverse, replies to those questions can be created.
 - Content creation - helps with creating blogs, reports or marketing text.
 - Data analyses - create analyses, evaluate insights, or classify data.
 - Workflow automation - schedule, report and enter data automatically.
- Prompt columns are licensed in the same way as AI Hub prompts. Other requirements include:
 - In the Environment's settings (in the Features area), Copilot and AI Prompts would need to be switched on.
- You can have up to 5 prompt columns per table.
- To create a prompt column:
 - create a column in a Dataverse table,
 - use the Data type "Prompt",
 - Deselect "Allow form fill assistance" checkbox,
 - If you don't want the prompt to be active, deselect the "Allow prompt column execution" checkbox,
 - Click "+Add new prompt",
 - Create a prompt,
 - Save the prompt,
 - Optionally add a filter,
 - If the filter is not met for a particular row, then there will be no prompt result for that row.
 - Save the column.
- Two additional columns will be automatically created, ending:
 - _PromptColumnStatus - either:
 - 0 - NotStarted,
 - 1 - InProgress,

11. Configure prompt columns

- 2 - Completed, or
 - 3 - Failed.
- `_PromptColumnDetails`
- After creating a prompt column, existing rows are not processed. However, they will be if an input column used in the prompt is updated.
- When creating a prompt, you may see a prefilled prompt.
 - To clear it, either manually delete it, or go to ... near Instructions and select "Clear prompt".
- Prompts should have:
 - Clear instructions,
 - Have sufficient detail, and
 - Be placed in sufficient context.
- There will already be an input column based on the primary column.
 - This is needed to save the prompt definition.
- To add an input column:
 - Click "+ Add content",
 - Select the relevant column in the dropdown list, then click Add.
 - Input columns can be added multiple times in the same prompt,
 - You cannot use formula columns, file columns, image data types, or other prompt columns.
- You can also click on the ... next to Instructions and go to Settings, where you can alter:
 - Temperature - a number between 0 and 1.
 - Lower temperatures will lead to more predictable and conservative outputs, and is best when you need high accuracy.
 - Higher temperatures allow for more diverse or creative and innovative responses.
 - The number of records retrieved for your knowledge sources,
 - Whether links will be included in the response to the relevant row,
 - Whether code interpreter is disabled, and

12. Configure row summaries

- The content moderation level
 - The default is Moderate.
 - Used for hate/fairness, sexual, violence or self-harm content.
- You can then test your prompt.
 - First, you need to select a relevant column.
 - Click on an input column,
 - Click on "Filter attribute",
 - Use an appropriate filter and enter a value.
 - Click "Close".
 - Then click "Test". The results will appear under "Model response".
 - You can check what data was used in the "Knowledge used" tab.
 - Finally, remove the filter, by clicking on:
 - An input column,
 - "Filter attribute",
 - "No filter",
 - Save.

12. Configure row summaries

- Row Summaries display a row's relevant details in a model-driven app.
- To be able to use them:
 - the "AI insights cards" needs to be switched on in the relevant environment by going to Settings - Product - Features - AI insights cards.
 - In the Power Platform Admin Center, Row summaries would need to be turned on in Copilot - Settings - Power Apps - Summary - Row summaries.
 - You then select either the Environment Group or the Environment.
- It is displayed:
 - At the top of the main form, after a row has been saved,
 - In a view, after clicking on the Summary button near a row.
- To edit how the row summary is generated:

13. Configure public views

- Go to the relevant table in Dataverse,
- Click on "Row summary" (in the Customizations section),
- In the Instructions section, click on "Add" or type / to display the list of columns.
- Select the relevant columns, and click "Add".
- You can also specify:
 - formatting - for example, in list form or in two paragraphs,
 - other instructions, if you have a preferred language.
- When viewing the row summary, you can:
 - Copy the summary in markdown format,
 - Use thumbs up/down to provide feedback,
 - In the main form, click on the Refresh button to regenerate it (using the latest data), and
 - Click Expand/Collapse to change the height of the summary.
- You can show/hide row summaries by:
 - going to the relevant table in Dataverse,
 - click on Forms (in the Data experiences section), and
 - clicking on Row summary - Show/Hide on all main forms.

13. Configure public views

- To create/edit a Public view, or edit a System view, go to:
 - Power Apps – Tables – select a table – Views.
 - Power Apps – Solutions – [NameOfSolution] – select a table (or add it to the solution) – Views
 - In the App Designer
 - In the classic Solution Explorer
 - From an app, Settings – Advanced Settings – Settings – Customizations – Customize the System – Tables – select the table – Views.
- There are three different types of views:

13. Configure public views

- Personal views (see topic 17)
- System views.
 - These can be edited but not deleted or deactivated, or shown in the view selector list, or used as sublists in a form or a list in a dashboard.
 - There are 4 types:
 - Associated – related tables for a row.
 - Advanced Find – display results when using Advanced Find.
 - Public views which include Contains/does not contain data are not included in the Advanced Find.
 - Lookup – select a row for a lookup column, and
 - Quick Find – default view for Quick Find. Also defines the columns that are searched when using Quick Find and Lookup view searches.
- Public views:
 - View called “Active” and “Inactive NAMEOFTABLE” will probably be created when you create a new table.
 - You can create public views and delete any public views you create.
- To edit system and public views:
 - Go to Power Apps – Tables – [NameOfTable] – Views, and:
 - Go to “New View”, or
 - Select an existing view and click Edit (or click on the ... next to it and go to Edit – Edit or Edit – “Edit in new tab”).
 - Click on the view (or right-hand click and go to “New Window”).
 - To view the “Table columns” pane, click “+ View column” in the top-left hand corner or near top-right, or the hamburger icon in the left-hand side and then “Table columns”.
 - In the filter dropdown, you can select All, Standard and Custom.
 - Any columns which have already been used will not be included in this list.

14. Configure main forms

- To add columns, click on it.
- You can also click on “Related” for related tables’ fields – tables which have a 1:N relationship.
- You can filter and sort in the right-hand pane.
- To remove columns, click on it and select “Remove”.
 - You can also sort, filter (using the expression builder), resize column widths, and move the columns there as well.
- To finish, click Save then Publish.
 - Note: this does not close the window.

14. Configure main forms

- To create a form – go to:
 - Power Apps – Tables – select a table – Forms or
 - Power Apps – Solutions – NameOfSolution – NameOfTable – Forms
 - It uses the Unified Interface, which is better for responsive design (as opposed to the Classic design).
 - To create a new form, go to “New form”.
 - To edit an existing form, click on the ... next to the form, and go to Edit – Edit or Edit – Edit in new tab.
 - You can also use the ... to “Turn off”, adjust the “Form settings”, “Delete” or see “Advanced – Show dependencies”.
 - In the properties on the right-hand side, you can adjust the:
 - Title,
 - Description (optional),
 - Max Width (in pixels – the default is 1900), and
 - Primary Image (optional).
- The main form is the main interface for table data.
 - Each table has to have a designated fallback form, used if there is no specified form for a particular security role. To get into this, go to:
 - Power Apps – Tables – [Name of Table] – Forms – Form settings – Fallback forms, or

14. Configure main forms

- Power Apps – Solutions – [NameOfSolution] – [NameOfTable] – Forms - ... - Form settings – Fallback forms
- If you have access to multiple forms, a form selector (with an arrow indicator) is shown at the top of the screen.
- If marked as “inactive”, the form will not be used.
- The tree view shows existing fields – you can also get to this by clicking on “Form field”.
 - Drag a field into position to add it into the body, or header/footer (a reduced field list).
 - You can hide the label, hide the field, either initially (it can be shown using code) or on the phone view, or make it read-only.
 - Click on the trash can to remove a field.
 - You cannot delete locked fields.
- The “table columns” shows all columns, or unused table columns.
- You can also add components, such as:
 - Layout
 - 1, 2 or 3 column tabs (on main forms only), or 1-4 column sections, or a spacer (a blank vertical space).
 - Grid
 - Editable grid or subgrid (a view of a related table).
 - Subgrids cannot be added to Quick Create forms.
 - Display
 - Calendar
 - Canvas app,
 - External website (also known as an iFrame)
 - It cannot be added to Quick Create forms
 - Form,
 - HTML web resource,
 - Image web resource (also shown under the Media heading),
 - Knowledge search,

14. Configure main forms

- Map
 - Needs enabling in your environment settings,
 - Only 1 map allowed per form,
 - Cannot be added to Quick Create forms
 - Also shown under the Media heading.
- Timeline.
- Input
 - Business card reader
 - Uses AI Builder.
 - Also shown under the AI Builder heading.
 - Checkbox,
 - Form,
 - Number/pen input,
 - Rich Text Editor Control,
 - Star Rating (from 1 to 5 stars), and
 - Toggle (on/off).
- Media
 - Image web resource and map (as per display).
- You can click on a fields and components and use the Properties pane to adjust whether they are:
 - hidden (so they are only shown through code),
 - hidden on phone
 - locked (not removed through the designer unless unlocked), or
 - read-only.
- You can also add Table columns into a Header:
 - Additional Table columns can be added into a flyout (arrow), if not hidden.
- To finish, click Save then Publish.
 - Note: this does not close the window. Click “Back” to go back.

15. Configure security

- You can create a custom security role by going to:
 - Power Apps – Admin Center – Environment – Access (right) See all – New role – enter a name.
 - Or share an model-driven app, and click on “New security role”
 - Locate tables, and add permissions:
 - Read, Create, Delete, Write,
 - Assign (give ownership of a record to another),
 - Share (give access of a record to another),
 - Append (attach an activity/note to a record), and
 - Append To (attached a record to an activity/note)
- You may have user or team privileges:
 - User privilege: create your own records, for instance.
 - Team privileges: create or view records with the Team (not individuals) being the owner, for example.
- Predefined security roles without a Dataverse database:
 - Environment Maker
 - Create new resources in an environment, including apps, connections, custom APIs, gateways and flows.
 - Can distribute their apps to other users.
 - Has no rights to access data.
 - Environment Admin
 - Can provision a Dataverse database.
 - View and manage all resources created in an environment.
 - Set DLP (data loss prevention) policies.
 - Add/remove users from Environment Admin/Maker role.
- Additional predefined security roles with a Dataverse database:
 - System Administrator

15. Configure security

- Full permissions to customize/administer the environment, including creating, modifying, and assigning security roles.
 - When a Dataverse database has been provisioned, all Environment Admin users will be assigned this System Administrator role.
 - Can view all data.
- System Customizer
 - Customize the environment (e.g. add new tables), but only has access to its own records.
- Basic User
 - Run an app and read, create, write and delete own records (standard tables).
- Delegate
 - Impersonate another user.
- If a table is custom, then you will need to define security when you publish.
 - Publish – Manage security roles
- Security role privileges are cumulative (additive): you have all privileges in all roles assigned to you.
 - User – access to your own records, records that shared with you, and records in which you are a team member.
 - Business Unit – User plus access to information throughout the business unit (usually reserved for managers).
 - Parent: Child Business Units – Business Unit plus all subordinate business units (not just their own). Usually for higher-level managers.
 - Global – All records in the organisation.
- These security roles can be assigned to:
 - users,
 - owner team and
 - Azure Active Directory group team
- An owner team and Azure AD group team can own records and has security roles assigned to the team.

15. Configure security

- They have full access rights to their records.
- The Azure AD group types are Security and Office.
- To create an owner team:
 - Go to Settings – Advanced Settings.
 - Go to System – Security – Teams, and click on “New”.
 - Enter team name, business unit (or the root business unit – the first one created for an organization), administrator, Owner, and any other fields.
- Adding a user to an environment
 - Power Apps – Admin Center – Environment – Settings – Users + Permissions – Users, and then
 - Add user if it doesn’t exist.
 - enter users and click “Add Users” button.
 - They are given:
 - Basic/Dataverse User, and
 - Environment Maker.
 - Removing users from a security group is done form the Microsoft 365 Admin Center - <https://admin.microsoft.com/>
- When creating model-driven apps, you need to add users to a security role that is assigned to the app.
- You can also configure column-level security.
 - Edit the appropriate column and check "Enable column security".
 - System Administrators will always have access to every column.
- You can then assign users to a column security profile, and give rights to that column.
 - Go to the Admin Center - Manage - Environments - [Name of Environment] - Settings - User + permissions - Column security profiles.
 - You can then add a new Column security profile.
 - Afterwards, you can add teams and users to that profile, and also add Allowed/Not Allowed permissions for Read, Read Unmasked, Update and Create.

Create model-driven apps

16. Create and configure forms for model-driven apps

- There are 4 different types of table forms.
 - Main (see topic 14)
 - QuickViewForm
 - Used only in related tables. Display additional data in related tables.
 - Needs a lookup field in the table (or the QuickViewForm will not be visible).
 - QuickCreateForm
 - Creating new rows.
 - Only the first available (based on security) Quick Create form per table will be used.
 - You can change the order by going to Tables – Forms and clicking on the ... next to the QuickViewForm.
 - Card.
 - Compact form view, good for mobile devices and dashboards. Can be color coded.

17. Create and configure views for model-driven apps

- Users with User level access to actions for the Saved View table can also create personal views.
- Owned by individuals, visible only:
 - by that user or
 - who that user share their personal views with.
- They can be created in the model-driven app by using:
 - Edit columns,
 - Edit filters. You can:
 - “add rows” (additional filters),
 - “add groups” (so you can use AND and OR between different filters), and

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18. Create generative pages by using natural language

- “add related entity” (filters from related tables).
- Go to the drop-down list next to the name of the current view and go to “Save as new view” or “Save changes to current view”.
- You can also “Set as default view”.
- You can also “Manage and share views”. This allows you to:
 - Hide the view,
 - Set as default view,
 - Share,
 - Edit info (name and description),
 - Assign,
 - Delete,
 - Deactivate, and
 - Download FetchXML.
- Personal views will have a head and shoulders icon next to them.

18. Create generative pages by using natural language

- You can create pages in model-driven apps, by describing what you want, and allowing Power Platform to build it.
 - At the time of writing, you need an environment in the US region, Great Britain, Australia or Singapore.
- To do this:
 - Open an existing model-driven app.
 - In the app designer, go to New – Describe a page.
 - Enter a description of the page (up to 50,000 characters – only US English is supported).
 - Include functional and optionally any User Experience (UX) requirements:
 - For example: Build a page showing the various employees. Include Level at the top, and beneath it Employee Name, Start Date and Salary.
 - Click the + at the bottom-left hand corner to:

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- link up to 6 Dataverse tables (no other data sources are allowed)
 - While most commonly-used data types are allowed, some others are not.
 - Optionally, add image(s) which shows what you intend the page to look like.
 - This can be a rough handwritten sketch.
 - This should not be used to add image(s) that should be used in the page. Instead, you can reference the image in a URL, a Dataverse table, or a web resource.
 - Optionally enabled/disable the Include images tool.
 - This allows the agent to include images from its library of 25,000 stock images.
 - These can be used as default/placeholder images, backgrounds, static content blocks, and more.
 - Optionally choose the AI model to be used.
 - Click the arrow to generate the page.
- The agent then goes through this process:
 - Step 1: Interpret the Request – the agent’s understanding of your prompt,
 - Step 2: Identify Requirements: Data Source, Fields to display, Layout, Responsible and Scrollable
 - Step 3: Assumptions – import the first few rows.
 - Step 4: Plan the Solution
 - Step 5: Verify Compliance
 - Code generation – the underlying code
 - Transpilation – ensures compatibility and proper rendering.
 - Final rendering.
- It is possible that you may need to click the Preview tab to see the final version.
- You can also manually edit the React code by clicking on the Code tab.
- You can chat with the app agent.

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- You can also attach a screenshot to help with your prompts to help adjust the visual.
- If you have multiple (two or more) iterations with the agent, you can click on “Compare” to view the differences between the current and previous version.
 - You can click on the “Accessibility assistant” to scan the code for accessibility issues.
- Finally, you can click on:
 - Save to save your progress, or
 - Save and Publish to publish the page.
- Note:
 - The following data types are supported:
 - Text
 - Text
 - Multiline Text
 - Number
 - Whole Number
 - Decimal Number
 - Floating Point Number
 - Date and time
 - Date and Time
 - Date Only
 - Lookup
 - Lookup
 - Currency
 - Choice
 - Choice
 - Yes/No
 - Other
 - Customer

19. Compose a model-driven app

- Image
- Status
- Status Reason
- Unique Identifier

19. Compose a model-driven app

- Model-driven apps design phases:
 - Model your business data
 - Table, Column, Relationship, Choice column (single or multiple values)
 - Security, including:
 - National/state/local data regulations
 - Company/trade regulations
 - User interface
 - App (components, properties, settings)
 - Pages and Navigation including:
 - Form (data entry),
 - View (a list of records).
 - Define your business processes / logic
 - Business rules – column requirements, hide/show columns, validate data.
 - Business process flow – how to go through submission. Does the user need approval for an invoice?
 - Workflow – automation not requiring Power Automate.
 - Actions – manual processes
 - Flows – automated workflows using Power Automate.
 - Build the app (output)
 - Charts (graphs)
 - Dashboards – one or multiple visuals
 - Embedded Power BI – tiles and dashboards

19. Compose a model-driven app

- Design it in the App Designer.
- In the App Designer:
 - Click on “Add page” (or go to Pages – New), and then select:
 - Dataverse table. This adds a connected form and view based on a Dataverse table.
 - If you click on the view, you can go to properties on the right-hand side, and select which Views and Charts can be used.
 - You can also rename the title of the view by going the “Settings” tab.
 - Dashboard.
 - In the Properties pane, you can select the Default Dashboard, the Title and Icon.
 - Navigation link – this is a website link. Web resource – a file created by a developer.
 - Custom page – this is a canvas app based on one screen.
 - In the “Advanced settings” or the “Settings” tab, you can also adjust the:
 - Privileges (whether a page is shown based on a user’s security),
 - Locale (translated titles and descriptions),
 - SKU (the versions of Dynamics 365 which display this page), and
 - Client (the clients which display this page).
 - All of the pages are shown on the left-hand side of the model-driven app.
 - If you want to edit a page, go to Pages – Navigation, hover over the page and click on the pencil icon.
 - You can also click on the ... and go to:
 - Edit table,
 - Edit command bar,
 - Move up (or down), and

20. Configure access to forms and views for model-driven apps

- Remove from navigation. This moves it to “All other Pages”. You can then click on the ... and go to:
 - “Add to navigation” (to move it back) or
 - “Remove from app” (which deletes it).
- Pages are separated into groups.
 - To add a new group, go to an existing group, click on the ... next to it, and select “New group”.
 - You can also select “New page”, “Move up” (or down), or “Remove from navigation”.
- You can also add areas in your model-driven app. This allows you to switch between sets of groups and pages.
 - This needs enabling by going to Settings – Navigation, or Pages – Navigation and going to the Properties pane.
 - You can select “Enable Areas”.
 - You can also unselect “Show Home”, “Show Recent” and “Show Pinned”.
 - You can also select “Enable collapsible groups”. This allows for groups to have a drop-down arrow next to them, and they can be collapsed, so that all of the pages are temporarily hidden.
- You can also add comments at the top of the App Designer.
- Once you have designed your model-driven app, you should go to the top of the app and click on Save, then Publish and then Play it to test it.

20. Configure access to forms and views for model-driven apps

- To configure access to a form or a view, click the ... next to the form (in Dataverse - tables) and click on "View/Form settings".
 - This means that you can create different views/forms (with access to different columns for different audiences).
- In the "Security roles" tab for a form or a system view, you can select:
 - Everyone, or
 - Specific security roles. Only these security roles will have access to this form/system view.
 - This does not apply to the default public view.

21. Configure access to model-driven apps

- You can select a view as the default view by clicking on the ... next to it and selecting "Set as default view".
- This is the only tab for views.
- In the "Form order" tab, you can select the order of the forms by dragging the forms up/down.
 - This is only used when a person has access to more than one form.
- In the "Fallback forms" tab, you can select the main form which will be used if a user does not have access to any forms.
 - This is because they are not part of any of the security roles assigned to the forms, and so probably should contain a minimum of data.
 - This is only used if there are multiple forms for a table.
 - For views, the default public view is always accessible.
- You can use the "Form access checker" tab to check what forms are enabled for a particular Security Role or App.

21. Configure access to model-driven apps

- Go to the Power Apps portal – Apps – click on the ... and go to Share.
- Select the app, then choose a security role.
- You can select “Manage security roles” to create a new custom role.
 - You then click “New”, then in the new window enter the “Role Name”, then select the relevant security.
 - You can locate tables, and add permissions:
 - Read, Create, Delete, Write,
 - Assign (give ownership of a record to another),
 - Share (give access of a record to another),
 - Append (attach an activity/note to a record), and
 - Append To (attached a record to an activity/note)
 - You can change them to:
 - None selected,
 - User,
 - Business Unit,

22. Create and configure model-driven app charts and dashboards

- Parent: Child Business Units, and
 - Organization.
- To use a model-driven app, a user will need to have:
 - privileges to the relevant Dataverse tables - at least read access
 - relevant security roles, and
 - access to the environment.

22. Create and configure model-driven app charts and dashboards

- Go to Power Apps – Tables – [NameOfTable] – Charts, and:
 - Go to “New chart”, or
 - Select an existing view and click Edit (or click on the ... next to it and go to Edit).
- You may need to blow up the window.
- You can create/edit charts from the App Designer or from Dataverse – Tables.
 - Microsoft Chart Controls are column, bar, area, line, pie, funnel, tag, doughnuts, multi-series, stacked and 100% stacked charts.
- You need to add a name, then which fields you are going to be using.
- There are personal charts, and system charts.
- There are also organization and user-owned charts.
 - Organization charts.
 - They are available across the organization.
 - There is no more detailed access level.
 - Need the System Administrator or Customizer role to create these.
 - User-owned visualizations
 - You have more control of access privileges and security.
- If you click on the ... next to a chart, you can:
 - Refresh,
 - Save As,
 - New (create new chart),

22. Create and configure model-driven app charts and dashboards

- Import/Export, and
- Move right/left.
- If you click on the ... next to a personal chart, then you can also:
 - Edit,
 - Share,
 - Assign (change the owner), and
 - Delete.
- You can create a public single-stream dashboard in the Power Apps portal – Tables – [Name of table] – Dashboards.
 - “Single-stream” means that it relates to a single table.
 - You can create a 2, 3 or 4-column overview, or a 3-column overview with a wider middle column.
 - You can access public dashboards by going to Open Dashboards at the top of the model-driven app (it may be hidden by the ...)
 - You can insert up to 2, 3 or 4 visual filters.
 - They display as charts, providing aggregated calculations.
 - They act as filters on a single tile – you can click on a specific bar, and the streams filter accordingly.
 - Visual filters are hidden in the dashboard by default. You need to click on “Show Visual Filter” in the dashboard’s command bar to display them.
 - You can also add multiple streams.
 - These can be based on views or queues. (Queues are used, for example, in Dynamics 365.)
 - You can configure what is shown in the view by customising the card form.
- You can then create a private multi-stream dashboard by going to a public single-stream dashboard and going to New – Dynamics 365 Dashboard.
 - “Multi-stream” means that it can relate to multiple tables.
 - You can choose from:
 - a 2, 3 or 4-column regular dashboard,

23. Create a canvas app using data

- a 3-column overview, focused, or multi-focused dashboard, or
 - a 4-column overview dashboard (this includes a list on the bottom row).
- You can insert Charts, Lists, Web resources, Timeline and iFrames (external website).
- You can also add Dashboards in the model-driven App Designer.
 - You go to Pages – “+ Add” – Dashboard, and add:
 - System dashboards,
 - Interactive dashboards, or
 - Power BI dashboards.
 - There are two types of dashboards: user and system dashboards.
 - An admin or customizer creates/customizes system dashboards for everyone in the organization.

Create canvas apps

23. Create a canvas app using data

- To create a canvas app using data, on the Home page, either:
 - click on Start with data, or
 - go to Create (on the left-hand side) and go to the "Start from data" section.
- You can select:
 - Create new data. This creates tables using in Data Workspace.
 - SharePoint list,
 - Dataverse table,
 - Excel Online (Business),
 - SQL database, or
 - Upload a CSV or Excel file.
- You can then select the relevant data and create an app.
- Alternatively, you can click on Apps (on the left-hand side), and go to "New app" - "start with data".

24a. Design apps for accessibility

- You can then create the tables in the Data Workspace and then click on "Save and open app".
- There is a drop-down next to it, so you can choose whether to create a Canvas app (the default) or a model-driven app.

24a. Design apps for accessibility

- Design for accessibility;
 - TabIndex allows for keyboard navigation.
 - It should mirror visual order – down or across.
 - Containers can group related content together.
 - Form Card and Gallery controls are automatically grouped.
 - Only have a tab stop at interactive controls.
 - Recommend values are:
 - TabIndex = 0 for interactive controls.
 - TabIndex = -1 cannot be accessed by keyboard navigation.
 - TabIndex > 0 can manually control the navigation order, but it is not recommend – it can break screen readers. Tip: “Check the order of the screen items”.
 - Visible=False or DisplayMode=Disabled means that the control is not included in keyboard navigation; use Size = 0 or Size =1 or change to Color to transparent if you don’t want it visible, but still want it available for accessibility.
 - Labels:
 - Live determines how screen headers interpret changes to label content.
 - Off – no changes announced.
 - Polite – waits until screen reader finishes.
 - Assertive – interrupts to say the changes.
 - Role
 - Is the Label a Heading? Heading1 to Heading 4, or Default for normal text.
- Validate for accessibility:

- You can use the App checker (a stethoscope in Power Apps Studio) or the Solution Checker to check for accessibility issues.
- AccessibleLabel is a label for screen readers.
 - Use in Image, Icon, and Shape controls.
 - Error in App Checker: “Missing accessible label”
- Adequate color contrast
 - between text and Fill or BackgroundImage.
- Screens
 - Screen readers will usually say the “Screen name” when loaded, as it is usually focused. (Use SetFocus immediately if you don’t want it spoken).
 - If screen has a default name, Tip: “Revise screen name”.
- Add picture buttons:
 - Must have Text and ChangePictureText for screen readers.
 - Focus indicators using FocusedBorderColor and FocusedBorderThickness
 - If FocusedBorderThickness = zero, error: “Focus isn’t showing”.
- Audio/Video controls:
 - If ClosedCaptionsURL is empty, Warning: “Missing captions”.
 - If default controls are turned off, Warning : “Missing helpful control settings”.
 - If Autostart is set to true, Warning: “Turn off autostart”.
- Pen controls:
 - If they exist, Tip: “Add another input method”
- Toggle or similar:
 - If the value labels (FalseText and TrueText) are turned off, Tip : “Add State indication text”.
- Generally:
 - If labels/markers for charts are not shown, Warning: “Missing helpful control settings”.

- If control contains HTML, and control is not an HTML text control, then Warning: “HTML won’t be accessible”.

24b. Design apps for performance

- Solution checker looks at Dataverse:
 - plugins and custom workflow activities
 - selecting all columns via Dataverse query APIs.
 - duplicate Dataverse plug-in registrations.
 - using batch request types in Dataverse plug-ins and workflow activities.
 - HTML and Javascript web resources
 - Don’t target absolute Dataverse endpoint URLs,
 - Don’t use modal dialogs,
 - Don’t use ‘eval’ or ‘alert’ functions.
 - configurations, such as SDK message steps, and
 - Canvas app formula issues, delayed load and whether there are too many app controls.
- It does not check the Default solution.
- To run the solution checker, go to Power Apps portal – Solutions – NameOfSolution - ... - Solution checker – Run.
 - The top-right of the screen says “Solution checker running”. It will take a few minutes.
 - You can cancel it while it runs, if you want.
 - You will receive an email and a notification when it is complete.
 - You can also view the results by going to Solution checker – View results.
 - Click an issue to view the details and how to resolve.
 - Columns include Issue [title], Category, Severity [high, medium, low and informational], Guidance, Component and Location.
 - You can download the results, by going to Solution checker – Download results.

- It downloads as a zip file.

24c. Design apps for responsiveness

- Responsiveness
 - Responsiveness is the ability to fit your app on the appropriate screen.
 - Your app could be viewed on a phone, tablet or desktop computer.
 - You should consider what sizes you want to support, and how the controls can be adapted.
 - Going to Settings - Display, you can:
 - Set the App layout to:
 - Fixed - the app dimensions are set by the designer,
 - Responsive - the app dimensions are set by the device.
 - If it is fixed, then you can:
 - Select portrait/landscape orientation,
 - You can lock/unlock the aspect ratio, and the lock orientation.
 - For tablets, you can select a screen size.
 - To assist with responsiveness, you can use the controls in the Layout section of the Insert menu, most notably containers.
 - You can insert items into those containers, and use formulas to calculate its position and size based on the size of the container.
 - You can also use combinations of horizontal and vertical container to help with auto-layouting the controls.
 - You can also show/hide items based on the screen size, by using the Visibility property and Parent.Size
 - If it is `>=ScreenSize.ExtraLarge`, it will typically be a desktop computer.
 - If it is `>=ScreenSize.Large`, it will typically be a tablet held horizontally.
 - If it is `>=ScreenSize.Medium`, it will typically be a tablet held vertically.

- If it is around `ScreenSize.Small`, then it will typically be a phone.

24d. Design apps for usability

- Usability
 - Screens
 - Ensure that the screens contain information which allow them to get from one screen to another, and to know which screen they are currently on.
 - Forms
 - Design your screen into areas which flow, and have related fields together.
 - Ideally, your form should be on a single screen. If not, design it to be on multiple screens, steps or tabs.
 - Add validation rules
 - Use the same form where possible for:
 - creating new records - the form should be in New mode.
 - editing existing records - the form should be in Edit mode.
 - displaying records - the form should be in View mode.
- Use containers to group together items (see topic 24c).
- Use reusable components (see topic 26).

25. Automate business processes from canvas apps

- Power fx functions that you can use to automate business processes include:
 - "Patch" - to create or update records directly, together with "Defaults" - the base records for a new record,
 - SubmitForm, NewForm, EditForm and ViewForm - to save a row, start to create a new record, and edit or view data in a form,
 - Set and UpdateContext - to assign values to global and context variables,
 - Navigate and Back - to move to other screens,
 - Collect, ClearCollect, and Clear to add, remove and add, and remove rows from a collection,

- If, IsBlank and IsEmpty to check for conditions and blank/empty states,
- IfError to run actions if there is an error in the action,
- Notify to notify the end user,
- FlowName.Run to run a Power Automate flow.

26a. Create reuseable components: named formulas

- You can create formulas and assign them to names in the App.Formulas property.
 - To select this, in the Tree View click on "App", and select Formulas.
- You can use the format:
 - `UserEmail = User().Email;`
- And then you can use this formula throughout your app.
- While you can create formulas in the App.OnStart property, there are advantages to using Named formulas in App.Formulas:
 - The formulas are always available.
 - They can refer to other Named formulas (but cannot create a circular reference), and do not have to be defined in any particular order.
 - The value does not need to be manually updated - it is always up to date.
 - The formula cannot be changed. However, the formula's value can change if its dependents change.
 - It is not calculated until needed, which can improve app loading time.
- You can also copy the Named formulas from one app to another.

26b. Create reuseable components: user-defined functions

- User-defined formulas can be defined in the App.Formulas property.
- The formula is defined as:
 - `FunctionName(`
 - `ParameterName1:DataType`
 - For example, `PriceBeforeTax:Number`
 - DataTypes can be:
 - Boolean - true/false,

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26c. Create reuseable components: component libraries

- Choice or Yes/No,
 - Color,
 - Currency (floating-point number),
 - Date, DateTime or Time
 - Decimal, Float or Number,
 - Dynamic - a variable data type,
 - GUID,
 - Hyperlink,
 - Image or Media (a URI text string),
 - Record (a row),
 - Record reference,
 - Table (containing multiple rows),
 - Text,
 - Void (no return type),
- You can have multiple parameters, separated by commas)
-): OutputDataType = Formula
- The Formula can include actions such as Set or Patch.

26c. Create reuseable components: component libraries

- You can create a standalone component in your canvas app by:
 - going to the Tree View (on the left-hand side),
 - clicking on the Components tab, and
 - clicking on "+ New component".
- However, it is restricted to your canvas app.
 - At the time of recording, there is a way to export/import components, but it is being retired, and Microsoft recommends not using this.
- Instead, in the Power Apps portal, you can go to More - Discover all, and click on "Component libraries".
- You can then create your component, including the size of the component.

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27a. Manage variables

- In the Properties pane of the component, you can add Custom properties, with:
 - Display name and name,
 - Description,
 - Property type (data, function, event or action),
 - Property definition (input or output), and
 - Data type.
 - You can also check "Raise OnReset when value changes" to reset the component.
- Any variables/collections created in the component are limited to the component.
- In an app, you can go to Insert (on the left-hand side) and click on "Get more components" (a folder icon) to import a component.
- If a component in a component library is updated, then:
 - App Maker are notified of these updated components, and are invited to review, and then update, the component.
 - Alternatively, if there is a component added into an app, in the Insert pane, you can click on the Refresh symbol next to the "Get more components" and click "Check for updates".

27a. Manage variables

- Use variables when you cannot use a reference to other objects and their properties.
 - For example, when you have a running total or the result of an “dialog box” (don’t know if that exists).
 - Or when it would slow down the computer by retrieving the same data – User(), as a whole record or one of its properties, for example, or querying a table.
- Global variables are assigned using the Set function:
 - Set(MyVar, 1) – set can be seen as the equivalent of “Let” in other languages.
 - Global variables can hold strings, numbers, records and tables.
 - Only one variable can be set at once using Set – for more, use ; (or ;; in some locales).

27b. Manage collections

- Because it is a “Let”, it does not create a link. Any changes to the original are not reflected in the variable.
- Variables do not need to be explicitly declared. Unassigned variables have the value of blank.
- It has a scope of the app – once the apps closed, it is forgotten.
- This is your default variable – only use Context variables if you are sure that’s what you want.
- They can also be used for input and output parameters to components.
- Context variables are only for one screen, so it’s good for passing variables to a screen – like parameters in a procedure.
 - You assign context variables using the UpdateContext or Navigate functions.
 - UpdateContext({ RunningTotal: RunningTotal + TextInput1 }, {IsSet: true})
 - Navigate(Screen1, None, { RunningTotal: -1000 })
 - More than 1 variable can be set at the same time.
 - Note – it is UpdateContext, not Update (which updates a record).
 - Because the scope is the screen, each screen can have the same variables – increases reusability.
- Both global and context variables can be viewed in Variables (on the left-hand side).

27b. Manage collections

- Collections are tables:
 - You can use them as if they were tables, but not with the Form control.
- Collect adds an item to a collection:
 - Collect(PaperTape, TextInput1.Text) – this is now in a column called “Value”
 - Collect(IceCream, { Flavor: "Pistachio", Quantity: 40 }, { Flavor: "Orange", Quantity: 200 })
 - Collect(myCollection, name_of_table)

28. Implement error handling

- But it is not delegable – only “500” (or 1-2,000) records will be retrieved.
- It is a copy of a table – not a link. Changes to the collection will not be reflected in the table.
- Clear clears a collection
 - Clear(PaperTape)
 - This leaves a blank collection, but the columns remain.
- ClearCollect does both
 - ClearCollect(IceCream, { Flavor: "Chocolate", Quantity: 100 }, { Flavor: "Vanilla", Quantity: 200 })
- You can also save and load the collection data to storage (note – not to a table):
 - SaveData(PaperTape, "StoredPaperTape")
 - Clear(PaperTape); LoadData(PaperTape, "StoredPaperTape", true)

28. Implement error handling

- Syntactical errors in formulas are highlighted when you write the error. They should be corrected when possible.
 - They can also be found using the App checker.
- You can check the values which are being handled by your formulas:
 - IsBlank allows you to check if the user input is empty - it returns true or value.
 - IsError returns true if the argument results in an error.
 - IfError returns the first argument, unless it results in an error, in which case it evaluates the next argument. (You can have multiple arguments.)
- If you are using the SubmitForm function, then use the OnFailure property for actions if it fails.
 - You can use the function Notify, which takes a string (error message) and a NotificationType.
- You can also use a more generic App.OnError property to do something if there is an error in the app.

29. Test canvas apps, including using Monitor

- You can test a canvas app by using Live monitor.

29. Test canvas apps, including using Monitor

- On the left-hand side of the app, go to Advanced Tools - Live monitor.
- You can then see what you do with the app in play mode. It records:
 - ID,
 - Time,
 - Category,
 - Operation,
 - Result and Result info,
 - Status,
 - Duration (in milliseconds),
 - Data source, and the
 - Control.
- To get more information about an action, click on it.
 - You can see a JSON version in the Details tab, and the formula in the Formula tab.
- You can also invite additional users by clicking on the "Invite" button.
- You can also:
 - Clear data,
 - Upload data, and
 - Download data.
- You can also go to Advanced Tools - Tests to create a suite of cases.
 - Test steps are a series of instructions or actions.
 - They are put together into a sequence in a “test case”.
 - They can be grouped together in “test suites”.
 - You can validate whether the test has run correctly using “test assertions”, a true/false expression. true means that it has run correctly; false means your test has failed.
 - Use: Assert(Condition, "Error message")
- Click “Record” and do your test.
- Then review the steps that have happened.

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30. Create a Copilot Studio agent from a canvas app

- Maybe alter one of the steps as a test.
- Then you can replay it.
- You can also click on the test, and select “Copy play link” to test it externally.
 - Different users can run the test on different machines.
 - You can then use the OnTestCaseComplete expression and the new variable TestCaseResult, with the properties:
 - StartTime, EndTime, TestCase- or TestSuite- Id, -Name, -Description (TestSuite as a series of partial test cases – if Case 1 ends on Screen2, Case2 will start there).
 - Success (true/false), TestFailureMessage (which step failed and why) and Traces (Assert and Trace entries).
- For maximum benefits from Test Studio:
 - Determine which test cases should be automated.
 - Repetitive tests.
 - High business impact functionality tests.
 - Features that are stable and not undergoing significant change.
 - Features that require multiple data sets.
 - Manual testing that takes significant time and effort.
 - Keep test cases small.
 - Makes it easier to isolate test failures.
 - Keep expressions to a single test action.
 - Every test case should have an expected result.
 - Use test suites.
 - For maintenance, group or categorize similar test cases together and describe the purpose and expected results of your test.

30. Create a Copilot Studio agent from a canvas app

- To create a Copilot Studio from an agent,
 - either:
 - Go to Apps (on the left-hand side),

31. Recommend cloud flow triggers

- check the canvas app, and
 - Go to "Create agent from app" (at the top).
- or:
 - Go to Agents (on the left-hand side) - you may need to go to More... first,
 - Click on "Create an agent from an app",
 - select an app.
- Then you should:
 - Define your agent - you may see a few suggestions - and click Next. You should:
 - Use specific wording,
 - Keep sentences simple. If they are more complex, divide them into separate sentences.
 - Ensure that they are in a logical order.
 - Review the agent's details, including the detailed instructions.
 - You can click on "Regenerate instructions" or edit them.
 - Click Create.
 - Introduction to cloud flows
- The first step in a cloud flow is a trigger.
- You can then insert actions, or parallel paths.

Create cloud flows

31. Recommend cloud flow triggers

- Automate workflows between your favorite applications and services, sync files, get notifications, collect data, and much more.
 - It is called "cloud flow" to distinguish it from "desktop flows" - there is a Microsoft Power Automate Desktop app used for this.
- Event driven flows are divided into:
 - Automated flows – triggered by an event: new email, Twitter posts, request for leave (approval flows).

32. Evaluate and recommend connectors

- Run when data is changed.
 - SharePoint, Dynamics, Outlook
- Scheduled flows – repeat every X minutes/hours/days/weeks
 - Run at a certain time, and reoccurs.
- Instant (button) flows – start with a tap of a button
 - Run when a virtual button is pressed on the mobile app.
 - Run when a physical button is clicked with 3rd party options.
 - Run when a button is pressed inside of Power Apps.
 - Allows you to run on demand.
 - Button trigger tokens include:
 - Location: Full address, Street, City, State, PostalCode, Country/Region, Latitude, Longitude,
 - Date/Time: Timestamp and Date, and
 - User: User name and user email
 - You can also “add text input” as well.
- There is a Power Automate mobile app.
- Connections are either:
 - Embedded (used in the flow), or
 - Other (defined, but not used in the flow).
 - Flows are shown in the Team flows tab – Invite another owner.

32. Evaluate and recommend connectors

- AI Builder
 - Types of actions and triggers – Standard
- Excel Online
 - Actions
 - Delete/get/update a row
 - Get worksheets/tables
 - List rows present in a table

32. Evaluate and recommend connectors

- Run script
 - Add a key column to a table (to the right)
 - Add a row into a table
 - Create table/worksheets
- Google Sheets
 - Actions
 - Get/insert/delete/update row
 - Get rows
 - Get sheets
- Microsoft Teams
 - Triggers
 - When a new channel message [not replies]/team member is added/removed
 - When I am mentioned in a channel message
 - For a selected message [button]
 - Sample actions
 - Add a member to a team
 - Create a Teams meeting/team
 - Get messages, post message/reply [in a chat/channel]
 - Post a choice of options as the Flow bot to a user
 - Post adaptive card in a chat/channel to a user
 - Reply with message/adaptive card
- MSN Weather
 - Triggers
 - When the current weather (temperature, UV Index, Humidity, Wind Speed) changes (e.g. wind “is equal to”, “goes over” or “goes below”).
 - When the current conditions changes (General)
 - Actions

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32. Evaluate and recommend connectors

- Get forecast for today/tomorrow
 - Get current weather
- Mail
 - send an email notification (action)
 - Not for spam! Maximum 5 mails per minutes, and 100 mails per 24 hours limit. Want more? Use Office 365 Outlook “Send an email”.
- Office 365 Outlook
 - Triggers
 - When a new email arrives [in a shared mailbox/mentions me]
 - When an email is flagged
 - When an event is created/added-updated-deleted/modified
 - When an upcoming event is starting soon
 - Actions
 - Create/delete/update contact/event
 - Send/delete/export/flag/forward/move/reply email
 - Find meeting times
 - Get attachment
 - Get calendar view of events
 - Get calendars/contact/contact folders/contacts/email/emails/event/events/mail tips for a mailbox/room lists/rooms/rooms in room list
 - Mark as read or unread
 - Respond to an event invite
 - Send an email from a shared mailbox
 - Send email with options
 - Set up automatic replies
 - Update my contact’s photo
- OneDrive for Business
 - Triggers

32. Evaluate and recommend connectors

- When a file is created/modified
 - For a selected file
- Actions
 - Create/delete file
 - List files in folder
 - Convert/copy/move or rename file [using path]
 - Create share link [by path]
 - Extract archive to folder
 - Find files in folder [by path]
 - Get file content/metadata [using path]
 - Get file thumbnail
 - List files in root folder
 - Update file [from URL]
- SharePoint
 - Triggers
 - When an item is created [or modified] [properties only]/is modified
 - When a file is created [or modified] in a folder
 - For a selected file/item
 - When a file is classified by a content understanding model
 - When a file/item is deleted
 - When a site has requested to join a hub site
 - Actions (some)
 - Hub sites
 - Approve/cancel hub site join request
 - Join hub site
 - Set hub site join status to pending
 - Add/delete attachment/get attachments
 - Check in/out file

32. Evaluate and recommend connectors

- Copy/move file/folder
- Create/delete/get/update file/item
- Create new folder
- Create sharing link for a file or folder
 - Stop sharing an item or a file
- Discard check out
- Get changes for an item/file (properties only)
- Get/update file properties
- Get files (properties only)/items
- Grant access to an item or a folder
- List folder
- Extract folder
- Microsoft Dataverse
 - Triggers
 - When a flow step is run from a business process flow
 - When a row is added, modified or deleted
 - When an action is performed (preview)
 - Actions
 - Add/delete/update row
 - Download/upload a file/image
 - Get a row by ID
 - List/relate/search/unrelate rows
 - Relate/unrelate uses 1:N or N:N relationships.
 - Perform a bound/unbound action
 - this means associated/not associated with a table
 - Predict using AI Builder models
 - Form Processing, Object Detection, Text Classification or use the prebuilt Business Card Reader, Key phrase

33. Manage approvals by using cloud flows

extraction, Language detection, Text recognition – OCR and Sentiment Analysis

- Perform a changeset request (a group of connector operations as a single transaction)
- SQL Server
 - Triggers
 - When an item is created/modified.
 - Actions
 - Delete/get/insert/update row
 - Execute a SQL query
 - Execute stored procedure
 - Get rows
 - Transform data using Power Query
 - Get tables

33. Manage approvals by using cloud flows

- You can create approvals to get human input for a cloud flow action.
- There are 5 approval types:
 - Approve/Reject - Everyone must approve,
 - Approve/Reject - First to respond
 - Custom Responses - Wait for all responses,
 - Custom Responses - Wait for one response,
 - Sequential Approval - approvals are requested in a specific order.
 - For example, one department needs to approve, and then another.
- To create an approval, add one of the following actions:
 - Create an approval
 - This does not wait for the approval to finish.
 - Start and wait for an approval
 - This pauses the flow until the approval/rejection has happened.

34. Configure cloud flow actions

- This has multiple parameters, including:
 - Title (required),
 - Assigned to (required),
 - Details, and
 - Link of the item to approve (and Description).
- Start and wait for an approval of text
 - This also include the "Suggested text" parameter.
- Wait for an approval.
- They create the "Response summary", "Completion date" and "Outcome" parameters.
 - You can then use the Control action to do different actions based on Outcome being Approve/Reject.
- The approver(s) will then receive an email, and can take the appropriate action.
- They can also see approvals by clicking on Approvals (on the left-hand side) in Power Automate. There are three tabs:
 - Received - approvals which have not been processed,
 - Sent - sent requests, and
 - History - past approvals and their Outcome.

34. Configure cloud flow actions

- Expressions
- Based on Azure Logic Apps.
- “?” in formula syntax indicates an optional input.
- String functions
 - concat('HI','YOU')
 - substring('hi there everyone', 6, 5) – zero-based
 - length('<text>')
 - replace('<text>', '<oldText>', '<newText>')
 - toLower, toUpper('<text>')
 - indexOf, lastIndexOf('<text>', '<searchText>')

34. Configure cloud flow actions

- Not case sensitive. Zero-based. If not found, returns -1.
- startsWith, endWith('<text>', '<searchText>')
 - true/false
- split('<text>', '<delimiter>') – returns array
- trim('<text>') – leading and trailing whitespace
- formatNumber(<number>, <format>, <locale>?)
 - Uses Excel Standard numeric format strings.
 - formatNumber(1234567890, '0,0.00', 'en-us')
 - C, D, E, F, G, N, P, R, X – see <https://docs.microsoft.com/en-us/dotnet/standard/base-types/standard-numeric-format-strings>
- Collections are for arrays
- Conversion include
 - array, binary, bool, float, int, string
- Logical
 - if(<expression>, <valueIfTrue>, <valueIfFalse>)
 - equals('<object1>', '<object2>')
 - and(<expression1>, <expression2>, ...), or, not(<expression>)
 - less, lessOrEquals, greater, greaterOrEquals(<value>, <compareTo>)
- Workflow gets inputs and outputs from flows
- Math
 - min, max(<number1>, <number2>, ...)
 - rand(<minValue>, <maxValue>)
 - add, sub, mul, div, mod(<d_1>, <d_2>)
 - Only two arguments. Can be nested.
 - Integer division results in integer.
 - range(<startIndex>, <count>)
 - range(1, 3) results in an array [1, 2, 3].
- Date and time

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35. Implement flow control with conditions and loops

- `utcNow('<format>')`
 - `utcNow()` - "2028-05-16T14:00:00.0000000Z"
 - `utcNow('D')` - "Tuesday, May 16, 2028"
- `getFutureTime, getPastTime(<interval>, <timeUnit>, <format>?)`
 - `getFutureTime(5, 'Day', 'D')`
- `addToTime, subtractFromTime('<timestamp>', <interval>, '<timeUnit>', '<format>')`
 - `addToTime('2028-05-14T14:00:00Z', 1, 'Day', 'D')`
- `addSeconds, addMinutes, addHours, addDays('<timestamp>', <seconds>, '<format>')`
 - `addSeconds('2028-05-14T13:00:00Z', -5)`
- `convertTimeZone, ToUtc, FromUtc, DateTime('<timestamp>', '<sourceTimeZone>', '<destinationTimeZone>', '<format>')`
 - `convertTimeZone('2028-05-14T14:00:00Z', 'UTC', 'Pacific Standard Time', 'D')`
- `startOfHour, Day, Month('<timestamp>', '<format>')`
 - `startOfHour('2028-05-14T13:00:00Z')`
- `dayOfWeek, Month, Year('<timestamp>')`
 - `dayOfWeek` returns 0 (Sunday) to 6 (Saturday).
 - `dayOfWeek('2028-05-14T13:00:00Z')`
- Variables
 - `variables('<variableName>')`
 - `variables('numItems')`

35. Implement flow control with conditions and loops

- Loops
 - Loop (not an actual action)
 - Runs an action until the end of the loop, then moves on to the next step in the flow.
- Switch

- What to run next, based on conditions from input.
- Do Until
 - Runs an action until a condition is TRUE.
- Apply to each.
 - Runs a set of actions for each item in an array.

36. Test and troubleshoot cloud flows

- Flow checker is always active in the command bar.
 - It displays a red dot when it finds errors, potential errors or warnings.
- Details are included in My flows - ... next to flow – Details.
 - You can click on the Flow Checker button to view the errors and warnings.
 - It can point to specific areas where you may wish to make improvements, together with guidance.
- Flows can fail due to:
 - Authentication failure
 - Action configuration issues
- You can Test the flow by clicking “Test”.
 - It will show the flow while running.
 - You can:
 - Perform the trigger action manually,
 - Use data from previous runs,
 - Use new data using connectors, such as Office 365 Outlook, SQL Server, Gmail and Outlook.com.
- Parallel paths are very useful with error handling.
 - You can click on the ... on an action and click “Configure run after” the previous action:
 - is successful,
 - has failed,
 - is skipped, and/or
 - has timed out.

- You can select multiple “run after” conditions for the same action.
- You can view the last flow runs by going to My flows and selecting the relevant flow.
 - This gives you a 28-day run history.
 - You can click on "All runs" for more runs.
 - You can also click on "Edit columns" to see more information about the runs.
- If a cloud flow fails, the flow owner will be sent an email, which includes:
 - Time
 - What happened - a description,
 - How to fix, and
 - Troubleshooting tips - this may include the count of the times the flow failed, and the ability to retry the flow with the same input data.
- These emails can be turned off in the flow's properties menu.
- Authentication errors:
 - A flow may fail due to an authentication error. This can be identified if you get:
 - An error message "Unauthorized",
 - An error code 401 "Unauthorized", or
 - An error code 403 "Forbidden".
 - In this case, you may need to fix the connection.
 - You will probably see a "Fix connection" link, which will enable you to verify your credentials.
 - You should then retest it by clicking on "Resubmit".
- If there is a problem within the flow itself,
 - then you may see:
 - An error message "Bad request" or error code 400, or
 - An error message "Not found" or error code 404.
 - In this case, you should edit the flow to correct the error.

37. Build prompts from templates or from blank

- If there is a temporary problem,
 - then you may see:
 - An error message 500 "Internal Server Error", or
 - An error message 502 "Bad gateway error".
 - In this case, just retry running the flow.
 - For error 500, you may need to wait 15-30 minutes first. If it continues look at [the Microsoft 365 Service Health dashboard](#).
 - For error 502, if it still happens, then check network and file settings, including any on-premises gateways.
- You can also use Copilot to troubleshoot problems.
 - It may not be able to solve all problems.

Create prompts and models in AI Hub

37. Build prompts from templates or from blank

- Prompts are instructions in natural language to ask a generative AI model to do something. You need:
 - AI Builder or Copilot credits,
- To create a prompt, go to the AI Hub (on the left-hand side), and then click Prompts.
- You can then create a prompt based on a template, or click on "Build your own prompt". Templates include:
 - Summarize/classify/extract information from/run sentiment analysis on text,
 - Respond to a complaint,
 - Anonymize/redact text,
 - Detect language,
 - Analyze word frequency, and
 - Identify text types
- You should provide instructions.
 - You can click "+ Add content" or type "/" to add:

38. Consume a prompt in apps

- A text input – you can name the input and add a sample text for testing,
- An image or document input – you can name the input and upload a sample image or document for testing,
- A Power Fx formula,
- A connection to Salesforce, Oracle Database or Zendesk,
- Data from a column in a Dataverse table.
- You can change the model.
- You can click on the ... to:
 - Clear the prompt,
 - Open the prompt library, or
 - Go to Settings. Here you can:
 - You can change the temperature.
 - Higher values will make the output more random.
 - Lower values will make it more focused and deterministic (more likely to be repeated in future iterations).
 - Say how many records will be retrieved,
 - Include links in the response,
 - Enable the code interpreter, and
 - Change the Content moderation level.
 - Medium is appropriate for most uses.
- You can test the prompt and see the output in Text, JSON or as a document.
 - After testing, at the bottom-right you can see how many Copilot credits it has cost.
- Before saving, you should name the prompt.

38. Consume a prompt in apps

- To use the prompt in a canvas Power App:

39. Consume a prompt in cloud flows

- Go to Data (on the left-hand side) - Add data - AI models - See all models - (Select the model).
- You can then use the formula 'NameOfModel'.Predict(Inputs).Text
- For example, you could add a button, and assign it to a variable use Set(NameOfVariable, 'NameOfModel'.Predict(Inputs).Text)

39. Consume a prompt in cloud flows

- To use the prompt in a cloud flow:
 - Click on + to add a new action,
 - Go to AI capabilities - Run a prompt.
 - In the Parameters tab, click on the Prompt dropdown, and select the prompt.
 - Then add any parameters which are needed.
 - You can then add Dynamic text in additional actions, and use Text (or Finish Reason or Data Used).

40. Add knowledge to a prompt

- See topic 37.

41. Customize prompt settings, including models

- See topic 37.

42. Add inputs to a prompt

- See topic 37.

Different AI models

- The different types of AI models include:
 - Both pre-built and custom models
 - Text Category Classification
 - Categorises text by meaning.
 - Entity extraction
 - Looks for location, dates, emails, names and numbers
 - Custom models

AB-410: Intelligent Applications Builder Associate
Different AI models

- Document processing model (previously known as Form processing)
 - Extracts text from images.
 - Works on JPG, PNG and unprotected PDF files.
 - Text-embedded PDFs as best.
 - Object detection model
 - Identifies objects from uploaded images
 - Counts number of objects included.
 - Objects on retail shelves (products densely packed)
 - Brand logo (logo detection)
 - Common objects (anything else)
 - Prediction model
 - Will something happen, based on past history data?
 - Binary prediction – Yes/No
 - Multiple outcome prediction – more than 2 outcomes.
 - Numerical prediction
- Prebuilt models:
 - Business card reader uses the form processing model
 - English only. Reads an uploaded photo/picture (jpg, png, bmp, tif).
 - If a business card, extracts required information.
 - ID reader
 - Passports and US driver licenses, social security cards and green cards. Does not retain image.
 - Invoice and receipt processing (2 separate models)
 - Invoice processing - English invoices, up to A3/Legal paper size.
 - Receipt processor - English receipts, such as restaurants, gas/petrol, retail receipts – not full page invoices. Up to 200 pages in a PDF.

43. Consume an AI model in apps

- Key phrase Extraction model
 - Trends from X, surveys, emails, forms.
 - Use in conjunction with sentiment analysis.
- Language detection model
 - Send received emails to language specialists
 - Save information in language-specific folders
 - “en” if English generally, “en-US” if US English, “unknown”
- Sentiment analysis
 - Positive/negative/neutral/mixed
 - Social media, customer feedback, email sentiment
- Text recognition model
 - OCR printed and handwritten text from images
 - car license plates, ID cards, handwritten notes

43. Consume an AI model in apps

- Power Apps
 - Go to Insert – AI Builder
 - Choose from: Business card reader, receipt processor, form (document) processor, object detector and text recognizer
 - To investigate the data, you can then go to:
 - Insert – Data table,
 - set Items to the name of the AI component, and
 - Edit fields – add fields.
 - Have a look at the Confidence – how sure is the AI that the data is correct?
- Formulas
 - Text recognizer properties (the Default properties can be set to these).
 - OriginalImage
 - Results. and Selected. – all text lines and the currently selected one. They have the following properties:

43. Consume an AI model in apps

- BoundingBox – coordinates
 - PageNumber
 - Text
- Receipt processor properties (all strings)
 - MerchantName, Address, Phone
 - TransactionDate, Time
 - PurchasedItems. – a list containing
 - Name, Price, Quantity and TotalPrice
 - Subtotal
 - Tax
 - Tip
 - Total
- Invoice Processing
 - Use code such as the following on a button:
 - `Set(FlowResults, Invoiceprocessingflow.Run(AddMediaButton1.Media)); Set(InvoiceId, FlowResults.invoice_id); Set(InvoiceTotal, FlowResults.invoice_total);`
- Business card reader:
 - AddressCity, Country, PostalCode, PostOfficeBox, State, Street
 - FullAddress
 - BusinessPhone, MobilePhone, Fax
 - CleanedImage and OriginalImage
 - CompanyName
 - Department
 - Email and Website
 - FirstName, LastName and FullName
 - JobTitle
- Sentiment Analysis

43. Consume an AI model in apps

- `AIBuilder.AnalyzeSentiment(TextInput1).sentiment`
 - `AIBuilder.AnalyzeSentiment("Esta fiesta es mi favorita", { language: "es-ES"}).sentiment`
- Category analysis
 - `Concat(AIBuilder.CategorizeText (TextInput1).categories , type & ";")` - Returns all categories that belong to the text, in list form.
- Language detection
 - `AIBuilder.DetectLanguage(TextInput1).language` - two-letter language code.
- Key phrase extraction
 - `Concat(AIBuilder.ExtractKeyPhrases(TextInput1).phrases, phrase & ";")` - key phrases.
- Entity extraction
 - `Concat(AIBuilder.ExtractTextEntities(TextInput1).entities, type & ", ")` - types of entities.
- Object detection properties
 - `ModelId` – which AI model?
 - `OriginalImage`
 - `GroupResults.`, containing
 - `TagID`
 - `TagName`
 - `ObjectCount`
 - `Results.` – the outputs, containing:
 - `BoundingBox`
 - `Confidence`
 - `TagId`, and
 - `TagName.`
- Document detection properties
 - `ModelId` – which AI model?

44. Consume an AI model in cloud flows

- OriginalImage
- Fields
- Tables
- Results., containing
 - BoundingBox
 - Confidence
 - PageNumber and
 - Value

44. Consume an AI model in cloud flows

- Add it into an existing flow, or search for “AI Builder” in the templates.
- Text
 - Analyze positive or negative sentiment
 - Classify text into categories with one of your custom models/standard model
 - Detect the language being used in text
 - Extract entities from text with one of your custom models/standard model
 - Extract the key phrases from text
- Objects
 - Detect and count objects in images
- Predict
 - Predict – custom AI builder models
 - Whether something will happen by field/record ID
- Process and save information from
 - forms, ID documents, invoices, receipts
- Read business card information
- Recognize text in an image
- Invoice Processing

45. Configure business rules

- Power Automate only – but can be called in Power Apps. Properties include:
 - invoice_id
 - invoice_total

Implement business and process logic

45. Configure business rules

- Business rules allow you to apply business logic at the data layer (not the app layer).
- They can, for both canvas and model-driven apps:
 - Validate stored data,
 - Set/clear field values,
 - Show error messages for invalid data.
 - This means that the data will not be saved.
- Model-driven apps only can also:
 - Show/hide/enable/disable fields,
 - In canvas apps, this would be require formula logic, not business rules.
 - Create business recommendations based on Business requirements.
- Business rules allow you to apply business logic at the data layer (not the app layer).
 - They can therefore be used in Power Apps, Power Automate, or API.
 - To access them, go to Tables – Business rules.
- Business rules have:
 - Condition – the trigger; should the “true” or “false” side of the business rule be run?
 - Action – logic in the true or false side.
 - Scope (for model-driven forms) – Entity/table, all model-driven Main and Quick Create forms, or a specific Main (not Quick Create) form
 - The scope for canvas apps is always “Entity”.

46. Configure business process flows

- All business rules apply when the form loads, and when the value of a relevant field is changed.
 - “Entity” business rules also apply when a record is saved, and when a record is created in Power Automate, APIs or the portal.
- To create a business rule, go to Tables, and then Business Rules.
 - Business rules run whenever a relevant field’s value is changed.
 - Once created, a business rule is Off by default.
 - You would need to click on it and click on "Turn on" for it to run.
- You cannot delete a business rule if it still on.
 - You need to turn it Off before deleting it.

46. Configure business process flows

- Streamlined user experience which guides the user through a process.
- Used with Model-driven Apps and the Dataverse (cannot use other data sources).
 - Can use up to 5 different tables in a single process.
 - Each table can have up to 10 business process flows.
- Define up to 30 stages, each with up to 30 steps, to get to a desired outcome.
- To create a Business Process flow:
 - go to Power Apps – Solutions – [Name of solution] – Objects - New – Automation – Process – Business Process Flow
 - go to Settings – Advanced Settings – Process Center - Processes – Processes – New.
- Click “Validate” at the beginning – this will add a message which will keep updating.
- The mini-map at the bottom left will help you navigate.
- Click “Snapshot” to download a png of the business process flow.
- After validation, you can then click Save (which turns into update) or Save As.
 - It is still a draft at this stage, and cannot be used.
- You can then click Activate.
- System business process flows included are:

46. Configure business process flows

- Lead to Opportunity Sales Process
- Opportunity Sales Process
- Phone to Case Process
- There are the following components:
 - Stage
 - Condition (If – true/false)
 - Can use AND or OR, but not both.
 - To branching back , use the Connector icon. All branches must branch back at the same time.
 - Steps
 - Can be marked “required”.
 - They need to be filled in the current stage; cannot save the form or move to the next stage without it – “stage-gating”
 - Data Step (add data)
 - If you have business-required or system-required columns, add these!
 - Workflow
 - Workflows can be triggered when going from one Stage to another on the Stage Exit.
 - You can drag workflows to the Global Workflow item in the designer (bottom of the screen) for:
 - Process applied
 - Process reactivated
 - Process abandoned (including moving to another business process flow) and
 - Process completed
 - If you want trigger a workflow at the beginning of the first stage, or the end of the last stage, then Stage Exit triggers will not work – you need to use a Global workflow.
 - Action Step

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47. Configure calculated, rollup, and formula columns

- Flow Step
- Make sure you click “Apply” at every stage or step, or your changes will not be kept.
- Run Business Process Flows in:
 - Power Apps: into a Model-driven App
 - Created from the App Designer – click on Business Processes – New. Embed into App.
 - Power Automate
 - an immersive business process flow.
 - Standalone – creates and uses its own table.
 - The WYSIWYG designer is not available for immersive business forms at this time.
- Work steps shown at the top of the App/flow.

47. Configure calculated, rollup, and formula columns

- When creating columns, enter Behavior:
 - Simple – a standard column
 - Calculated
 - uses columns from the current table or related parent tables.
 - Can use AND/OR operators,
 - Supported by text, choice, Yes/No, Whole/Decimal Number, Currency and DateTime column types.
 - Functions include ADD/SUBTRACT/DIFFIN-HOURS, -DAYS, -WEEKS, -MONTHS, -YEARS, CONCAT
 - TRIMLEFT, TRIMRIGHT removes not spaces, but the first/last X number of characters.
 - Use for:
 - Multiplying/adding/subtracting/dividing columns
 - Calculating total costs
 - Follow up dates (in 7 days’ time, for example).

47. Configure calculated, rollup, and formula columns

- Only recalculated when a row is saved, not when a field is updated.
- Rollup – up to 10 columns per table, and 100 per organization.
 - SUM, COUNT, MIN, MAX and AVG
 - Calculate totals, counts, earliest time.
 - Cannot trigger a workflow.
 - Not instantaneous – automatic, and the maximum recurrence setting is one hour.
 - It creates a `_date` column (when it was last recalculated) and a `_state` column, which contains:
 - NotCalculated,
 - Calculated,
 - OverflowError (too big),
 - OtherError (a temporary internal error),
 - RetryLimitExceeded (due to concurrency and locking conflicts),
 - HierarchicalRecursionLimitReached (due to too big hierarchy being used),
 - LoopDetected (recursive loop).
- You can also use the Formula data type:
 - Use Power fx functions (similar to Excel)
 - Don't start with an = sign.
 - It can create the following Formula data types: Decimal, Whole number, Floating point number, and Choice.
 - You can also select:
 - Searchable - to be available in views, charts, dashboards and Advanced Find, or
 - The number of decimal places.
 - The formula is calculated in near real-time

48. Evaluate use cases for business logic

- You may also be able to use a "Get formula suggestions" input, which would then create a Suggested Formula, which you can copy and use/edit.
 - This is currently available in environments in the US region.

48. Evaluate use cases for business logic

- Business use cases for business logic include:
 - Having default values for columns.
 - For example, you may have 2 columns called "State" and "Country". Using "Set Field Value", if you change "Country" to "United Kingdom", then "State" could be automatically changed to "Not applicable".
 - Also, you could "Set a default value" - for example, if no data is entered, then "United States" will be assumed.
 - To stop/allow data from being added
 - You could then prevent "State" for being changed, if the "Country" is the "United Kingdom".
 - Showing more detail only if it is necessary ("Set Visibility").
 - For example, if you have 3 BusinessAddress fields, you might not want to show BusinessAddress3 if BusinessAddress2 is blank.
 - To validate data
 - This allows data to be checked before it is saved.
 - To require data to be entered
 - You could set a field to "Business Required" or "Not Business Required (Optional)".
 - To provide guidance for a value
 - It can show a "Recommendation".